

Running TimesheetTracking

A start-to-finish runbook with commands, verification, troubleshooting, and cleanup.

ABOUT THIS GUIDE

A standalone beginner guide for the Team 3 TimesheetTracking Kubernetes exercise. Use it together with the other part of the field guide.

Namespace	timesheet-ns
Hostname	timesheettracking.local
Ingress	Traefik
Application	NGINX Alpine

Running TimesheetTracking

Field Guide - Part 1 of 2

A start-to-finish runbook for standing up the TimesheetTracking exercise on Rancher Desktop, with every command, every expected result, and every real problem hit along the way.

Namespace	timesheet-ns
Hostname	timesheettracking.local
Ingress class	traefik
Folder	exam\01\TimesheetTracking

Before you start

- **Rancher Desktop is installed and running** - its whale icon should be visible and settled, not spinning, in the Windows system tray.
- **Kubernetes is turned on inside Rancher Desktop** - Preferences -> Kubernetes -> Enable Kubernetes.
- **You have a Command Prompt window open** - regular Command Prompt is enough; nothing here needs Administrator rights except the two steps marked below.
- **kubectl responds** - Rancher Desktop installs it for you, you don't need a separate download.

Seven steps, in order

Each step depends on the one before it - the namespace has to exist before anything can live inside it, and the cluster has to be awake before it will accept any of this.

1. Create your workspace folder

Every file for this exercise - the five YAML files, nothing else - lives in one folder. Creating it first keeps the whole exercise self-contained and easy to find again later.

```
:: creates the folder, then moves into it
mkdir "D:\Projects\kubernetes\exam\01\TimesheetTracking"
cd /d "D:\Projects\kubernetes\exam\01\TimesheetTracking"
```

The `/d` switch just makes sure Command Prompt also switches drive letter, in case you started on `C:`.

2. Confirm the cluster is actually awake

Rancher Desktop's Kubernetes takes a little while to boot after a fresh start - running your files against a cluster that isn't ready yet is the single most common source of confusing errors, so we check first.

```

:: which cluster kubectl is pointed at
kubectl config current-context

:: is the one node ready
kubectl get nodes

:: is the Traefik ingress controller running
kubectl get pods -n kube-system | findstr traefik

:: is an IngressClass named traefik registered
kubectl get ingressclass

```

Expect: context `rancher-desktop`, one node `Ready`, a `traefik-...` pod showing `1/1 Running`, and an ingressclass named `traefik`.

If the node isn't ready yet - *Rancher Desktop can take several minutes to finish starting Kubernetes, especially the first time after a reboot. Wait and re-run `kubectl get nodes` every 30 seconds or so rather than assuming something is broken.*

3. Write the five YAML files

Namespace, ConfigMap, Deployment, Service, then Ingress - save each one in the folder from Step 1. What each field means is covered file-by-file in **Part 2: Understanding TimesheetTracking**; this guide only concerns itself with running them.

Filenames used throughout this guide: `01-namespace.yaml`, `02-configmap.yaml`, `03-deployment.yaml`, `04-service.yaml`, `05-ingress.yaml`.

4. Apply the files, in order

"Applying" a file tells Kubernetes: *make the real cluster match this description*. The namespace has to be created before anything can be placed inside it, so the numbered order matters - one command with several `-f` flags works as long as the namespace file comes first in the list.

```

kubectl apply -f 01-namespace.yaml -f 02-configmap.yaml -f 03-deployment.yaml -f 04-service.yaml -f 05-ingress.yaml

```

Expect: five lines, each ending in `created - namespace/timesheet-ns created`, `configmap/...created`, and so on.

5. Check that everything actually came up

Applying a file only submits a request - it doesn't guarantee the result worked. These checks confirm the whole chain is connected end to end.

```

:: pod, deployment, service all in one view
kubectl get all -n timesheet-ns

:: does the Service have a real pod behind it
kubectl get endpoints -n timesheet-ns

:: does the Ingress have an address assigned
kubectl get ingress -n timesheet-ns

```

Expect: the pod shows `1/1 Running`, the endpoints list is **not empty** (an IP address, not blank), and the Ingress `ADDRESS` column is **not blank** - a blank address means Traefik hasn't picked up the rule yet.

6. Point your computer at the hostname

Kubernetes now knows about `timesheettracking.local` - but Windows doesn't, yet. The hosts file is the short list Windows checks before asking the internet how to find a name, and this step adds one line to it. It's the one step here that needs an **elevated (Administrator)** Command Prompt.

```
:: run from an Administrator Command Prompt
echo 127.0.0.1 timesheettracking.local >> C:\Windows\System32\drivers\etc\hosts
```

Prefer a text editor instead? Right-click Notepad -> "Run as administrator", open `C:\Windows\System32\drivers\etc\hosts`, and add the line `127.0.0.1 timesheettracking.local` at the bottom.

7. Test it

```
curl http://timesheettracking.local/web
curl http://timesheettracking.local/status
```

Expect: TimesheetTracking web service is running and TimesheetTracking service is healthy.

Getting a Windows / IIS 404 page instead? See Incident 01 below - something else on your machine already owns port 80.

Incident log

Three problems worth knowing about - real issues encountered while building this exact exercise on Windows with Rancher Desktop, not hypothetical edge cases.

Symptom	Cause	Fix
01 - curl returns an IIS "HTTP Error 404" page, not your text, even though the Ingress ADDRESS looked fine.	Windows' built-in web server (IIS, service name <code>W3SVC</code>) was already listening on port 80, so Rancher Desktop's own port-forwarding couldn't bind to it - every request to port 80 was quietly being answered by IIS instead of Traefik.	Confirm with <code>Get-NetTCPConnection -LocalPort 80</code> in PowerShell, then stop IIS with <code>net stop W3SVC</code> from an Administrator prompt. Restart it later with <code>net start W3SVC</code> if you need it for something else.
02 - kubectl says "actively refused" right after starting Rancher Desktop.	The Kubernetes control plane genuinely hadn't finished starting yet - this is normal, especially on the first start after a reboot, and can take several minutes.	Wait, then re-run <code>kubectl get nodes</code> . No fix needed beyond patience.
03 - hosts file "Access is denied" when adding the line.	The system hosts file is protected; a normal Command Prompt or PowerShell window doesn't have permission to write to it.	Reopen Command Prompt (or Notepad) with "Run as administrator", then retry the same command.

Cleanup

Deleting the namespace removes everything inside it in one shot - the ConfigMap, Deployment, Pod, and Service. The Ingress is also namespaced, so it goes with it; only the hosts-file line is outside the cluster and needs removing separately.

```
kubectl delete namespace timesheet-ns
```

Then remove the `timesheettracking.local` line from the hosts file the same way you added it (Administrator Notepad), and run `net start W3SVC` if you stopped IIS in Incident 01 and want it back.

Copy-paste reference

```
:: 1 - folder
mkdir "D:\Projects\kubernates\exam\01\TimesheetTracking"
cd /d "D:\Projects\kubernates\exam\01\TimesheetTracking"

:: 2 - cluster check
kubectl config current-context
kubectl get nodes
kubectl get pods -n kube-system | findstr traefik
kubectl get ingressclass

:: 4 - apply
kubectl apply -f 01-namespace.yaml -f 02-configmap.yaml -f 03-deployment.yaml -f 04-service.yaml -f 05-ingress.yaml

:: 5 - verify
kubectl get all -n timesheet-ns
kubectl get endpoints -n timesheet-ns
kubectl get ingress -n timesheet-ns

:: 6 - hosts file (Administrator prompt)
echo 127.0.0.1 timesheettracking.local >> C:\Windows\System32\drivers\etc\hosts

:: 7 - test
curl http://timesheettracking.local/web
curl http://timesheettracking.local/status

:: cleanup
kubectl delete namespace timesheet-ns
```

*Part 1 of 2 - TimesheetTracking Field Guide. See **Part 2: Understanding TimesheetTracking (CONCEPTS-GUIDE.md)** for what each file and each Kubernetes idea actually means.*

Live version: <https://claude.ai/code/artifact/28da4d10-1fc3-46ac-9617-b765ba86c6fd>